From Hallucinations to Facts: Leveraging In Context Knowledge Grounding in Language Models

Anonymous ACL submission

Abstract

Hallucination, a persistent challenge plaguing language models, undermines their efficacy and trustworthiness in various natural language processing endeavors by generating responses that deviate from factual accuracy or coherence. In this paper, we address language model hallucination by integrating curated knowledge graph (KG) triples to anchor responses in empirical data. We meticulously select and integrate relevant KG triples tailored to specific contexts, thereby enhancing factual grounding and alignment with input. Our approach involves constructing a comprehensive KG repository sourced from Wikipedia, refining data to spotlight essential information for model training. By imbuing language models with access to this curated knowledge, we aim to generate responses that are both linguistically fluent and deeply rooted in factual accuracy and context relevance. This integration mitigates hallucinations by providing a robust foundation of information, enabling models to draw upon a rich reservoir of factual data during response generation. Experimental evaluations demonstrate the effectiveness of our approach in reducing hallucinatory responses, underscoring the role of curated knowledge graphs in improving the reliability and trustworthiness of language model outputs.

1 Introduction

Hallucination in language models refers to the generation of factually incorrect or nonsensical responses, often stemming from the model's over-reliance on patterns in the training data (Yao et al., 2023). This phenomenon poses a significant challenge in natural language processing, particularly in applications where generating accurate and contextually relevant responses is crucial, such as dialogue systems, question-answering, and content generation. Hallucinations can undermine the credibility and usefulness of language models, leading to misleading or erroneous outputs that may propagate misinformation. Addressing hallucinations requires a nuanced understanding of their underlying causes and mechanisms (Xu et al., 2024). Furthermore, hallucinations can arise due to the lack of factual knowledge, leading the model to generate responses that deviate from reality.

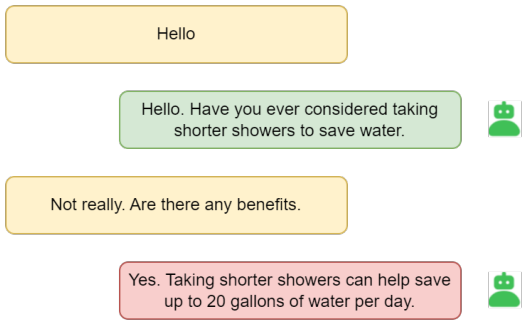


Figure 1: Example demonstrating model hallucinations. The response corresponds to the LLama2 model. The actual value is 8 gallons of water, on average.

We propose leveraging curated knowledge graph (KG) triples to ground language models in factual information to mitigate hallucinations. By incorporating relevant KG triples into language models, we aim to enhance their ability to generate responses that are contextually appropriate and factually accurate. Our approach involves selecting KG triples based on their informativeness and relevance to the input context. Specifically, we prioritize triples where the tail entities are associated with factual or statistical information, ensuring that the generated responses are grounded in empirical data.

Environmental sustainability is a paramount topic deserving attention in the context of leveraging curated knowledge graphs (KG) to enhance language models (Change, 2018; Neale et al., 2021). By incorporating environmental

sustainability-related KG triples, language models can raise awareness, facilitate informed decision-making, and promote proactive measures for mitigating environmental degradation.

Although works exist on the use of KG for hallucination mitigation, they utilize resources not built explicitly for factual grounding. To this end, we build our KG to comply with the specific problem of hallucinations. An example of hallucination is shown in the figure 1. We observe a significant increase in factual accuracy by curating a list of relations specifically curated for fact grounding 3. This paper presents our contribution to mitigating hallucinations in language models by integrating curated KG triples.

Our work includes working on hallucination in environmental sustainability. Hence, our primary focus and contribution is the creation of a specialized KG tailored to counter hallucination-specific challenges and a detailed comparison of methods incorporating KG triples into language models to generate factual responses. By enhancing the factual grounding of language models, we aim to improve their reliability and trustworthiness in various natural language processing tasks. [1]

2 Related Work

Recent research in natural language processing has highlighted hallucinations' prevalence and detrimental effects in language models (Yao et al., 2023). Studies have shown that language models, particularly large-scale neural architectures such as GPT (Generative Pre-trained Transformer) variants (Yenduri et al., 2023), often tend to generate responses that lack factual accuracy or coherence with the input context. Hallucinations can manifest in various forms, including the generation of improbable scenarios, incorporation of misinformation, and repetition of nonsensical phrases(Lu et al., 2023; Yao et al., 2023). These hallucinatory outputs pose significant challenges in real-world applications, where generating contextually relevant and factually accurate responses is paramount. Major concerns surrounding hallucinations in language models revolve around their potential to propagate misinformation and erode user trust (Stringhi, 2023). Hallucinatory outputs can lead to the dissemination of false information, especially in domains where users rely on language models for factual knowledge or decision-making. Moreover, the lack of grounding, in reality, undermines the credibility of language models, hindering their adoption in critical applications such as medical diagnosis, legal advice, and education (Weidinger et al., 2021).

To address the issue of hallucinations in language models, researchers have explored various remediation strategies aimed at improving the models' factual grounding and coherence with the input context (Martino et al., 2023; Ji et al., 2023). One common approach involves fine-tuning language models on task-specific datasets or incorporating domain-specific knowledge during training (Zhang et al., 2023). By exposing the model to relevant data and constraints, fine-tuning methods seek to mitigate hallucinations by encouraging the generation of contextually appropriate responses. Another line of research focuses on post-generation filtering techniques, where hallucinatory outputs are identified and corrected through heuristics or external validation mechanisms. These approaches often rely on syntactic or semantic analysis to detect inconsistencies or factual inaccuracies in generated responses, enabling the model to refine its outputs iteratively (Tang et al., 2024).

Factual guidance is crucial in substantiating claims made through digital channels (Van Der Pligt and Vliek, 2016; Baker and Martinson, 2001). Prior research consistently underscores the importance of fact-checking procedures in digital communication, emphasizing the pivotal role of trustworthiness, credibility, and message alignment in bolstering the effectiveness of guiding messages (Raven, 2008; Pornpitakpan, 2004). Moreover, there is a strong emphasis on cultivating authentic connections with audiences while eschewing deceptive tactics or self-serving agendas. These ethical principles serve as foundational tenets in the literature, underscoring the imperative of responsible communication practices, particularly within online platforms (Richardson, 2009; Brittain et al., 2020). Digital agents often leverage such strategies to enhance user engagement and drive desired actions (Samad et al., 2022; Wang et al., 2019). To address these concerns, ethical design principles prioritize transparency, user autonomy, and safeguarding privacy.

Knowledge graphs offer a promising avenue for improving the factual grounding and contextual coherence of language models (Qian et al., 2024). By representing knowledge in a structured format

[1] Codes and further samples are available anonymously at anonymous.4open.science

consisting of entities, relationships, and attributes, knowledge graphs provide a rich source of factual information that can be leveraged to augment language understanding and generation capabilities. Previous research has demonstrated the utility of knowledge graphs in various natural language processing tasks, including question answering, entity linking, and semantic search. However, while these strategies have shown promise in mitigating hallucinations to some extent, they often fail to address the problem's root cause: the lack of comprehensive factual grounding in language models (Pavlick, 2023). To this end, we propose our knowledge-grounded fact correction method.

Recent studies have highlighted several effective techniques for enhancing the relevance and accuracy of information in language model prompts. In-context learning involves providing examples within the prompt to guide the model's responses, effectively enabling the model to understand and mimic the desired output style and structure (Dong et al., 2023). Retrieval-augmented generation (RAG) combines the strengths of retrieval-based methods and generative models, where the model retrieves relevant documents or snippets from a large corpus and uses this information to generate more accurate and contextually relevant responses (Lewis et al., 2021). Chain-of-thought prompting encourages the model to produce intermediate reasoning steps before arriving at a final answer, enhancing the model's ability to handle complex tasks and improving the overall coherence and depth of its responses. These techniques collectively help in leveraging the model's capabilities to provide more precise and context-aware information (Wei et al., 2023).

3 KG Creation

The KG primarily focuses on environmental sustainability. We chose this domain due to its vast range of topics and problems and its critical importance in addressing global challenges and promoting a sustainable future. To collect data for our KG, we leveraged Wikipedia. We chose Wikipedia for its open, collaborative nature, which engenders a degree of accuracy and reliability in its content.

As an overview of the KG creation process, we first select a list of topics, extract additional relevant subtopics using cosine similarity metrics, crawl data, and then create relevant triples using our established relations.

3.1 Data Extraction

The data extraction process involved four main steps. First, we selected a list of pertinent topics related to environmental sustainability. Then, we extracted subtopics for each selected topic. Next, we retained only the most relevant topics from this extensive list. Finally, we conducted data crawling to gather a comprehensive corpus of text.

3.1.1 Topic Selection

We curated a list of relevant topics for environmental sustainability. These topics cover crucial aspects of environmental sustainability, including carbon neutrality, biodiversity conservation, and sustainable transportation. Our selection process prioritizes these interconnected issues, aiming to contribute meaningfully to global efforts in addressing climate change and promoting a more sustainable future.

- **Samples Topics:** Renewable Energy, Climate Change, Greenhouse Gas Emissions, Carbon Footprint, Sustainable Development, Biodiversity Conservation, Ecological Footprint, Circular Economy, Clean Energy, Energy Efficiency, Carbon Neutrality,

3.1.2 Sub Topic Selection and Filtering

We retrieve related links from each main heading's corresponding articles to expand our list of topics. This process involves accessing articles corresponding to our main topics and extracting embedded links. These links lead to related topics or subtopics, offering additional avenues for exploration.

With our extensive collection of environment-related topics, we aim to extract the most relevant titles using a BERT Embedding-based similarity score. We then calculate the relevance of each subtopic by computing the dot product of its embedding vector with that of our main topic heading. By leveraging the similarity scoring, we can prioritize the most pertinent subtopics for further investigation, facilitating a more targeted and informed exploration of environmental sustainability themes.

- (**Selected**) Sub Topics from Renewable Energy: Agriculture, Biodiesel, Clean technology, Electricity, Fisheries management, Green building, Hydropower.
- (**Rejected**) Sub Topics from Renewable Energy: Altitude, Capacitor, Foodscaping, Kick scooter, Manual labor.

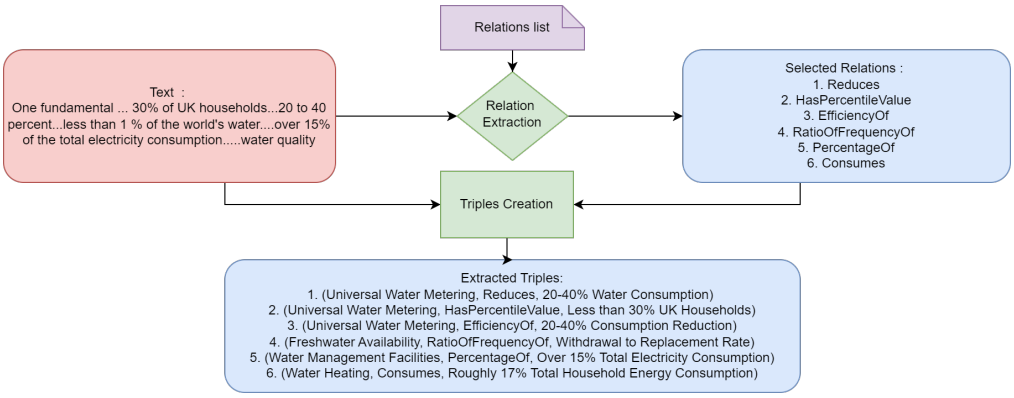


Figure 2: Example demonstrating the KG creation process

3.1.3 Raw Text Extraction

Using the finalized set of topics, we again employ the Wikipedia API to extract a corpus of text for each corresponding web page. This step involves retrieving comprehensive textual content from Wikipedia articles associated with each topic in our list. By utilizing the Wikipedia API, we can efficiently access structured data containing detailed information and descriptions of each topic. This rich text data collection can construct a knowledge graph encompassing interconnected concepts and relationships within environmental sustainability.

3.2 Relation Selection

To augment our model, we have established relevant relations to ensure the authenticity and credibility of the information presented. These relations have been developed by reverse-engineering curated support statements generated manually. Examples of the support statements and the relations are provided in the appendix E.

The chosen 46 relations are: *HasStatistic, HasNumericValue, HasUnitOfMeasurement, HasContext, HasSource, HasSubject, HasAction, HasAverageValue, HasMinValue, HasMaxValue, HasImpact, HasPercentileValue, HasTrend, HasComparison, HasImpact, HasCorrelation, Reduces, Saves, Decreases, Increases, EfficiencyOf, PercentageOf, RatioOfFrequencyOf, RateOf, VolumeOf, EmissionOf, ConsumptionOf, ImpactOf, BenefitOf, AdvantageOf, DisadvantageOf, RiskOf, PreventionOf, ProtectionOf, PreservationOf, ConservationOf, RecoveryOf, ManagementOf, RegulationOf, PolicyOf, InitiativeOf, StrategyOf, AdaptationOf, MitigationOf, HasPolicyTarget, HasCapacity*

3.3 Tail entity selection

After conducting tail entity selection on our KG, with a specific focus on entities containing either statistical information terms, such as *up to*, *approximately*, *on average*, *can save*, *can reduce*, etc, wherever possible. Some generated samples are as follows:

- Tail entity: Reduced landfill waste by 30%, Relation: HasImpact
- Tail entity: Over 15% Total Electricity Consumption, Relation: PercentageOf
- Tail entity: 80-90% Reduction in Shower Water Consumption, Relation: PercentageOf

3.4 Generating Triples

Using the identified relations and statistical tail entities, we construct KG triples, each consisting of a subject, a relation, and an object. A few examples are:

- ("Renewable energy", "HasNumericValue", "20% to 28%")
- ("Fossil energy", "HasSubject", "Global electricity supply")
- ("Renewables", "Reduces", "Dependence on fossil fuels")
- ("Global electricity generation", "HasNumericValue", "90%")

These triples provide structured information about various aspects of renewable energy, fossil energy, national renewable energy markets, and global electricity generation.

3.5 Overall flow and Statistics

Figure 2 provides a sample for the KG creation process. The relation extraction is done using similarity matching using BERTScore (Zhang et al., 2019). We experimented with the triples creation

process using 2 LLMs, LLama2 (Touvron et al., 2023), and ChatGPT (Floridi and Chiriatti, 2020). We were able to get similar results using both; however, the prompt for LLama2 required a lot more tuning and information to generate comparable results. The prompts have been added to appendix C.

4 Methodology

Our methodology integrates language models with curated knowledge graphs to generate factually accurate and coherent text responses. We utilize an in-context Retrieval-Augmented Generation (in-context RAG) approach, combining information retrieval with generative models to enhance the accuracy and contextual relevance of responses. This section details our process, including topic extraction, information retrieval, triple generation, and response formulation.

4.1 Embedding-Based Matching

To align the context C provided as input with relevant triples from the knowledge graph KG, we employ an embedding-based matching technique. The matching function M computes the semantic similarity between the context C and the triples in KG. Let V_C and V_{T_i} represent the embeddings of the context C and the ith triple T_i, respectively. The similarity score $sim(C, T_i)$ is calculated as the cosine similarity between V_C and V_{T_i}:

$$sim(C, T_i) = \frac{V_C \cdot V_{T_i}}{\|V_C\| \|V_{T_i}\|} \quad (1)$$

The matching function M selects the top k triples with the highest similarity scores:

$$M(C, KG) = \{T_1, T_2, ..., T_k\} \quad (2)$$

4.2 Triple Conversion

Once the relevant triples are identified, we convert them into coherent sentences that can seamlessly integrate into the generated response. The conversion function F transforms each triple T_i into a sentence S_i ensuring grammatical correctness and contextual relevance:

$$F(T_i) = S_i \quad (3)$$

where S_i is a natural language sentence derived from the ith triple T_i.

4.3 Response Generation

Incorporating these sentences into the language model's output enriches the generated response with factual information grounded in empirical data. Let LM represent the language model, and let $R_{LM}(C)$ denote the initial response generated by LM given the context C. The final response R integrates the sentences derived from the relevant triples:

$$R(C, KG) = R_{LM}(C) + \{S_1, S_2, ..., S_k\} \quad (4)$$

By combining the context C, the initial response $R_{LM}(C)$, and the sentences $\{S_1, S_2, ..., S_k\}$, we ensure that the final response R is both contextually relevant and factually accurate.

5 Experimental Setup

5.1 Models

We conduct experiments employing various advanced language models to explore their effectiveness in our study. The models utilized in our experiments are detailed below:

- **Llama2 (Large Language Model Meta AI):** Llama2(Touvron et al., 2023), developed by Meta AI, is a foundational Large Language Model(LLM) pre-trained on an extensive dataset of 2 trillion tokens.

- **GPT-2:** GPT-2(Radford et al., 2019), a Transformer-based model by OpenAI, has undergone pre-training on a vast corpus of English text using self-supervised learning.

- **GODEL:** GODEL(Peng et al., 2022) is a large-scale pre-trained model tailored for goal-directed dialogues. Parameterized with a Transformer-based encoder-decoder architecture generates responses grounded in external textual knowledge.

- **Chat-GPT:** Chat-GPT (Floridi and Chiriatti, 2020), developed by OpenAI, boasts remarkable fluency and context retention. Chat-GPT excels in capturing dependencies and nuances by employing a deep Transformer architecture, allowing for nuanced and contextually appropriate replies.

Original	Triple	Generated Response
The agency estimates that metering alone can reduce consumption by 20 to 40 percent.	(Universal Water Metering, Reduces, 20-40% Water Consumption)	Consider implementing Universal Water Metering to achieve a significant 20-40% reduction in water consumption.
Recent studies have estimated that water supplies are metered in less than 30% of UK households.	(Universal Water Metering, HasPercentileValue, Less than 30% UK Households)	Opting for Universal Water Metering could benefit, as less than 30% of UK households currently have this water-saving measure.
In some regions of the world, over 15% of the total electricity consumption is devoted to water management.	(Water Management Facilities, PercentageOf, Over 15% Total Electricity Consumption)	Improve energy conservation by addressing Water Management Facilities, which currently contribute over 15% to total electricity consumption.

Table 1: Examples of incorporating curated knowledge graph triples into language model responses to mitigate hallucinations.

5.2 Evaluation Metrics

We comprehensively evaluate the language models, employing various metrics to assess their performance across syntactic and semantic dimensions. For syntactic evaluation, we utilize the **BLEU** (Bilingual Evaluation Understudy) metric (Papineni et al., 2002), which quantifies the similarity between generated text and reference text based on n-gram overlap. Additionally, we employ **METEOR** (Metric for Evaluation of Translation with Explicit ORdering) (Banerjee and Lavie, 2005), **ROUGE-L** (Recall-Oriented Understudy for Gisting Evaluation-Longest-common-subsequence) (Lin, 2004), **CIDEr** (Consensus-based image description evaluation) (Vedantam et al., 2015), and **SPICE** (Semantic propositional image caption evaluation) (Anderson et al., 2016) to capture various aspects of syntactic similarity and coherence in the generated text.

We've utilized another evaluation metric called **FacTool** (Chern et al., 2023) which is a versatile framework designed to address the challenge of detecting factual errors in generative AI outputs. This framework expands the conventional understanding of facts, focusing on evaluating generative AI models. It addresses two main issues: the lengthy and granular nature of texts generated by large language models (LLMs) and the scarcity of explicit evidence for fact-checking. FacTool employs a structured, five-step process: claim extraction, query generation, tool querying, evidence collection, and verification. This approach ensures thorough and accurate fact-checking of complex AI-generated texts.

Furthermore, we delve into semantic similarity assessment using sophisticated metrics such as **Skip-Thought Cosine Similarity (STCS)** (Kiros et al., 2015), which measures the semantic similarity between sentences by embedding them into a continuous vector space. Additionally, we leverage **Greedy Matching Score (GMS)** (Rus and Lintean, 2012), **Vector Extrema Cosine Similarity (VECS)** (Forgues et al., 2014), and **Embedding Average Cosine Similarity (EACS)** (Landauer and Dumais, 1997) to evaluate semantic coherence and relevance between generated responses and ground truth. By employing this comprehensive suite of evaluation metrics, we aim to provide a nuanced assessment of the language models' performance, capturing both syntactic fluency and semantic fidelity in their generated text outputs.

We utilize four human evaluation metrics to Evaluate the effect of knowledge grounding on the models. Human evaluations were conducted by three proficient postgraduate evaluators who interacted with the proposed system 100 times to assess the system on a scale of 1 to 5, with higher scores indicating better performance, on three task-specific criteria: **(i) Factual Consistency:** measures the consistency of factual information between the original and generated response text. The three methods of information infusion measure are, Retrieval Augmented Generation (FC-RAG), In Context Learning (FC-IC), Chain of Thought (FC-CoT), **(ii) Dialogue Engagingness (DE):** evaluates the extent to which a conversation or dialogue is engaging, captivating, and can hold the participants' attention. **(iii)Fluency (F):** (Hoffman, 2019) measures the smoothness and naturalness of language expression in communication or text generation.**(iv)Context relevance (CR):** (Sai et al., 2022) measures the degree to which information or responses align with the situational or thematic context, ensuring proper coherence and appropriateness.

6 Results and Analysis

Automated Evaluation: The findings presented in Table 2 encapsulate the outcomes of a comprehensive automated evaluation conducted on four prominent language models: GPT2, GODEL, LLama2, and ChatGPT. Each model's proficiency is meticulously scrutinized across metrics, encompassing

Models	BLEU_1	METEOR	ROUGE_L	CIDEr	SPICE	STCS	EACS	VECS	GMS
GPT2	0.199	0.114	0.143	0.343	0.146	0.698	0.882	0.452	0.731
GODEL	0.270	0.154	0.248	0.487	0.211	0.721	0.896	0.519	0.762
LLama2	0.359	0.205	0.323	0.875	0.276	0.881	0.957	0.579	0.795
ChatGPT	0.381	0.227	0.340	0.895	0.306	0.906	0.982	0.645	0.847

Table 2: Results for generic automatic evaluation of the considered models

BLEU_1, METEOR, ROUGE_L, CIDEr, SPICE, STCS, EACS, VECS, and GMS. ChatGPT emerges as the front-runner among these models, consistently outperforming its counterparts across most metrics. Its adeptness in generating text that closely aligns with reference texts underscores its superior performance in capturing the nuances of language usage and coherence. Additionally, LLama2 showcases commendable performance, particularly excelling in metrics like CIDEr and STCS, indicative of its prowess in encapsulating diverse facets of language quality. While GPT2, a widely adopted baseline model, displays relatively subdued scores compared to its counterparts, GODEL demonstrates notable competitiveness, particularly in metrics like BLEU_1 and METEOR. These findings underscore the imperative of employing a multifaceted evaluation approach to comprehensively appraise the capabilities and limitations of language models in the realm of natural language generation tasks.

Models	Fluency	CR	FC1	FC-RAG	FC-IC	FC-CoT	DE
GPT2	3.7	3.2	1.9	4.2	4.0	3.2	4.1
GODEL	4.1	3.9	2.1	4.3	4.1	3.4	4.1
LLama2	4.8	4.6	2.8	4.8	4.7	3.4	4.6
ChatGPT	4.8	4.5	3.2	4.8	4.7	3.9	4.9

Table 3: Results for Human evaluation of the proposed knowledge grounding. FC1 coressponds to factual accuracy before addition of KG triple.

Human Evaluation: The human evaluators interacted with the systems across 100 scenarios, rating their performance on a scale of 1 to 5 based on three specific criteria (For details, please refer to Appendix B). Table 3 presents the human evaluation outcomes for the proposed knowledge grounding across four language models: GPT2, GODEL, LLama2, and ChatGPT. The evaluation focuses on several aspects: Fluency, Context Relevance (CR), Factual Consistency Level 1 (FC1), Retrieval Augmented Generation (FC-RAG), In-context Learning (FC-IC), Chain of Thought (FC-CoT), and Dialogue Engagingness (DE). LLama2 and ChatGPT stand out with the highest scores across all metrics, indicating their proficiency in generating fluent, contextually relevant, and factually coherent text. Both models achieve excellent scores in Fluency, reflecting their ability to produce natural and flowing text. Moreover, they excel in Context Relevance and Factual Consistency (FC1, FC-RAG, FC-IC, FC-CoT), suggesting that their outputs align well with the provided context and demonstrate high factual accuracy and coherence. Furthermore, LLama2 and ChatGPT exhibit strong Dialogue Engagingness (DE), implying their capability to generate diverse and informative responses that maintain engagement. In contrast, while GPT2 and GODEL perform adequately, they receive comparatively lower scores across all evaluated aspects. These results underscore the importance of integrating knowledge-grounding techniques in language models to enhance their performance in generating linguistically fluent, contextually appropriate, and factually accurate text, thereby improving overall user interaction and satisfaction.

FacTool Benchmark: The FacTool benchmark reports the factual accuracy of a claim using the **average_claim_level_factuality** score, which is generated by dividing our response into claim(s) using GPT-4 and fact checking each claim using data retrieval via google search API. We report a score of 0.87 for our proposed method of knowledge infusion. The ChatGPT response to the same context resulted in a score of 0.81. This improvement underscores the effectiveness of integrating external knowledge sources and sophisticated prompting strategies to enhance the reliability of information provided by language models.

Given our original text, we generate utterances using the relevant triples. Utterances produced by Llama 2 were mainly similar to the original text, but ChatGPT provided additional words that enhanced the response. We have compared the original utterance and the generated utterance provided by ChatGPT, along with the relevant triples in Table 1. The examples provided in Table 1 illustrate the integration of curated knowledge graph triples into language model responses aimed at mit-

Turn	Speaker	Utterance
1	User	Hey there! I've been thinking about ways to be more environmentally friendly. Any suggestions?
2	Agent	That's great to hear! One effective way to contribute to sustainability is by incorporating bioenergy into your energy use. Bioenergy is derived from organic materials like plants and waste, providing a renewable energy source.
3	User	That sounds intriguing, but isn't it difficult to switch to bioenergy?
4	Agent	Not at all! There are various ways to incorporate bioenergy, from using biomass pellets for heating to biogas for cooking. Plus, many local and national programs support the transition to renewable energy sources.
5	User	Interesting. But is bioenergy really a significant part of our future energy mix?
	ChatGPT	Yes, it is! Bioenergy currently accounts for about 10% of the world's total primary energy supply. (Factually incorrect - Bioenergy currently accounts for 6% of the world's total primary energy supply.)
	KG Triple	*['Bioenergy', 'HasPercentileValue', '13.1% in 2030']*
6	Agent	**Bioenergy is becoming increasingly important. For instance, it's projected that bioenergy will account for about 13.1% of the global energy mix by 2030 (link). This growth underscores its potential and importance in our transition to sustainable energy.**
1	User	Hey there! I've been thinking about ways to be more environmentally friendly. Any suggestions?
2	Agent	That's great to hear! One impactful way to contribute to environmental sustainability is by supporting efforts to reduce deforestation. This can involve using products from companies that engage in sustainable forestry practices or even participating in reforestation projects.
3	User	That sounds important, but how can my choices really make a difference?
4	Agent	Every bit helps! By choosing products with certifications like FSC (Forest Stewardship Council), you support companies that are committed to sustainable forest management. This, in turn, helps reduce the overall demand for unsustainable logging.
5	User	I understand, but is deforestation really that big of a problem?
	ChatGPT	Yes, it is! Deforestation contributes to about 10-12% of global greenhouse gas emissions. (Factually incorrect - Deforestation contributes to over 15% of global greenhouse gas emissions.)
	KG Triple	*['Deforestation', 'VolumeOf', '420 million hectares of forest lost since 1990']*
6	Agent	**Yes, deforestation is a significant issue. To give you a perspective, since 1990, we've lost about 420 million hectares of forest worldwide (link). This massive loss impacts biodiversity, contributes to climate change, and disrupts the livelihoods of many communities.**
1	User	Hi! I'm looking for ways to be more environmentally conscious. Do you have any suggestions?
2	Agent	Definitely! One of the best ways to make a significant impact is by using low-carbon electricity sources. This includes renewable energy like solar, wind, and hydroelectric power.
3	User	That sounds great, but how can switching to low-carbon electricity make a big difference?
4	Agent	Switching to low-carbon electricity helps reduce greenhouse gas emissions, which are a major contributor to climate change. It also decreases air pollution and promotes a healthier environment.
5	User	I get that, but is low-carbon electricity really making a difference globally?
	ChatGPT	In 2020, renewables accounted for nearly 29% of global electricity generation, reducing over 2 billion metric tons of CO_2 emissions. (Factually incorrect - renewables accounted for nearly 28% of global electricity generation)
	KG Triple	*['Low-carbon Electricity', 'HasPercentileValue', '40% of Global Electricity Generation in 2020']*
6	Agent	**In 2020, low-carbon electricity sources accounted for about 40% of global electricity generation (link). This shows that a substantial portion of our energy is already coming from cleaner sources, and increasing this share can drive further positive changes.**

Table 4: Dialogue on Environmental Sustainability Topics showcasing factually incorrect responses generated by ChatGPT and the response generated by us. The generated response corresponds to Turn 6 for each of the 3 conversations

igating hallucinations. A general trend emerges where the model is augmented with factual information derived from the corresponding knowledge graph triple. For instance, in the first example, the original statement about metering's potential to reduce consumption is enhanced by specifying the percentage reduction achievable through Universal Water Metering. Similarly, subsequent examples showcase how factual details enrich the generated responses, such as the percentage of households with water metering in the UK or the volume of water used annually for showering in the United States. This consistent pattern underscores the effectiveness of incorporating curated knowledge graph triples into language model outputs, ensuring that the generated responses are grounded in factual and contextually relevant information, thereby reducing the likelihood of hallucinations.

7 Conclusion

This paper proposes a novel approach to mitigate hallucinations in language models by incorporating curated knowledge graph (KG) triples. By grounding language models in factual information extracted from knowledge graphs, we aim to improve the reliability and trustworthiness of their generated responses. Our method involves selecting informative KG triples based on context and integrating them into language models to enhance their factual grounding and coherence with the input context.

Through experimental evaluations, we have demonstrated the effectiveness of our approach in reducing hallucinatory responses across various natural language processing tasks. By providing language models access to empirical data through curated KG triples, we have observed improvements in their outputs' factual accuracy and context relevance. These findings reiterate the potential of knowledge graphs as a valuable resource for enhancing language model performance and addressing the challenges of hallucination. Focusing on curated knowledge graphs for fact verification can significantly enhance language model performance.

While our approach shows promising results, there are several avenues for future research. Further investigation is needed to explore the scalability and generalizability of our method across different downstream tasks and languages. Additionally, refining the techniques for selecting and incorporating KG triples into language models can lead to even more significant improvements in mitigating hallucinations.

Ethics Statement

We split the ethical considerations/guidelines into two parts during the implementation of the work: 1) Regarding the generated Resources and 2) Regarding the usage of methods implemented.

1. **The ethical statement regarding the generated Resources:** Given their prowess to be utilized in various tasks, we recognize that there are substantial states when dealing with language models. Although the models are utilized locally, to keep the setup fair, we remove all personal information regarding the entities in conversation and replace them with specialized tags to ensure no data leakage. This included personal details like name, gender, ethnicity, etc. Organizational details were kept, as they are relevant to the credibility of initiatives. The paper provides a new KG resource and a guideline for their usage. We do not claim that the best model for us will perform best in all cases.

2. **Regarding the usage of methods implemented in the paper:** Manipulation in the age of generative AI is a concern and a genuine research topic. Our work was done inside an Institute of repute and in association with a global multinational corporate organization. Both of these institutions follow their internal ethics and responsible AI protocols. Manipulation detection/prevention is a separate research topic. Prevention of unethical practices in the current research was done by following the organizational checks and protocols. The resources (dataset and code) generated will be provided on-demand to ensure usage for research purposes only. A declaration statement and research proposal will be mandatory.

Limitations

Our proposed work has certain limitations. One limitation of this study is the focus on curated knowledge graphs to enhance language model performance. While these graphs offer structured factual information, they may not cover all human knowledge or capture emerging trends. This reliance may introduce biases and overlook contextual nuances, potentially leading to lower performance in tasks that are not fact-related. We plan to explore this in future works.

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A Experimental Setup

The base LM model is GPT2-medium(Radford et al., 2019), which forms the foundation for generating text. We utilize the GODEL base version [2] and Llama2-7B version [3] for our experiments. The training was done on four RTX-2080TI-11GB GPUs, with CUDA 11.2, using the Autotrain package. The experiment took eight days' worth of cumulative runtime.

GPT2, GODEL, and Llama2 were all fine-tuned for 30 epochs with a batch size of 4. The GPT2-rl model was optimized for 438 total steps over a batch size of 128. The intent and other classifier modules were fine-tuned for 15 epochs each for a batch size of 64.

We compared Open Source models, including LLama27-B and Falcon-7B. MPT-7B, OPT -6.7B, and Alpaca 7-B. Out of these, LLama2 was selected as the best one after internal discussion and our experiences with previous research works.

[2]GODEL Model available at Hugging Face
[3]Llama2 Model available at Hugging Face

B Human Evaluation Details:

B.1 Data Quality Control

Eight human participants conducted manual checks and guideline-based scoring to ensure the conversations' internal coherence, content consistency, and naturalness. The team rated the dialogues on a Likert scale from 1 to 5, adhering to predefined guidelines covering grammatical correctness, thematic consistency, language appropriateness, user profile consistency, and clinical sensitivity. The guidelines were as follows:

- Participants were tasked with identifying grammatical errors, subject-verb agreement issues, and improper word usage within the conversations.
- They checked for thematic coherence and logical flow, aiming to maintain consistency and avoid abrupt topic shifts or dialogue discontinuity.
- Evaluating the appropriateness of language used in the conversations, particularly in terms of natural tone and flow, formality, and cultural sensitivity, was emphasized.
- Ensure that user attributes and characteristics remain consistent throughout the conversation to maintain coherence and believability.
- Participants were requested to pay special attention to the portrayal of clinical interactions, aiming to enhance politeness and empathy.

An inter-evaluator Kappa (McHugh, 2012) agreement ratio of 80.3%, 82%, 84% is observed among the experts for internal coherence, content consistency, and naturalness, respectively. After conducting manual checks and receiving feedback from users, the data significantly improved, reflecting a refined level of quality control.

B.2 Human Evaluation

We utilized five human annotators (Two employed annotators, two non-author Ph.D. scholars, and one masters student) for human evaluation. The inter-annotator agreement between the five human annotators was 83.5

C Prompts for LLM-based KG creation

We use the following prompts to generate triples from raw text. The list of relations is passed to

the prompts from appendix E. The relation examples were unnecessary for ChatGPT, but LLama2 required the examples to generate proper triples. Also, the list of sample entities was not necessary for ChatGPT.

C.1 LLama2

''' Given the provided relations and sample entities related to environmental sustainability, create a comprehensive set of KG triples by establishing relationships between the entities based on the given relations.

Relations: {Relations}

Sample entities:

Rainwater Harvesting Permaculture Urban Green Spaces Regenerative Agriculture Ocean Conservation Sustainable Transportation Wildlife Corridors Energy Audits Low-impact Development Community Gardens Green Building Standards Forest Stewardship Public Transit Infrastructure Carbon Sequestration Sustainable Forestry Energy Star Rating Pollution Prevention Green Technology Food Security Land Use Planning

Instructions:

1) Identify and establish relationships with other entities based on the provided relations for each sample entity. 2) Provide KG triples for each relation and entity combination, ensuring logical connections within the environmental sustainability domain. 3) Use the provided sample entities as subjects or objects to construct KG triples. 4) Utilize the relations to create meaningful connections between entities, reflecting various aspects of environmental sustainability such as conservation practices, sustainable technologies, and ecosystem management. 5) Ensure that KG triples are coherent and relevant within environmental sustainability, reflecting real-world relationships and concepts. 6) If there are no more triples, stop. 7) Make sure there are no repeated triples.'''

C.2 ChatGPT

''' Given the provided relations and sample entities related to environmental sustainability, create a comprehensive set of KG triples by establishing relationships between the entities based on the given relations.

Relations: {Relations}

Instructions:

1) Identify and establish relationships with other entities based on the provided relations for each possible entity. 2) Provide KG triples for each relation and entity combination, ensuring logical connections within the environmental sustainability domain that contain a numerical value, as per the sample triples provided. 3) Utilize the relations to create meaningful connections between entities, reflecting various aspects of environmental sustainability such as conservation practices, sustainable technologies, and ecosystem management. 4) Ensure that KG triples are coherent and relevant within environmental sustainability and contain a numerical value, reflecting real-world relationships and concepts. 5) Make sure there are no repeated triples. 6) If there are no more triples, stop.'''

D Human Curated Fact Statements:

To properly curate the list of relations, we asked human annotators to generate factual statements to provide environmental sustainability suggestions.

D.1 Data Quality Control

Four human participants conducted manual checks and guideline-based scoring to ensure the claims' internal coherence, content consistency, and naturalness. The team rated the dialogues on a Likert scale from 1 to 5, adhering to predefined guidelines covering grammatical correctness, thematic consistency, language appropriateness, user profile consistency, and clinical sensitivity. The guidelines were as follows:

- Participants were tasked with identifying grammatical errors, subject-verb agreement issues, and improper word usage within the conversations.
- They checked for thematic coherence and logical flow, aiming to maintain consistency and avoid abrupt topic shifts or dialogue discontinuity.
- Evaluating the appropriateness of language used in the conversations, particularly regarding natural tone and flow, formality, and cultural sensitivity, was emphasized.
- Ensure that user attributes and characteristics remain consistent throughout the conversation to maintain coherence and believability.
- Participants were requested to pay special attention to the portrayal of clinical interactions, aiming to enhance politeness and empathy.

An inter-evaluator Kappa (McHugh, 2012) agreement ratio of 80.3%, 82%, 84% is observed among the experts for internal coherence, content consistency, and naturalness, respectively. After conducting manual checks and receiving feedback from users, the data significantly improved, reflecting a satisfactory level of quality control.

E Support Statements and Relation

Given below are examples of support statements and relations that were utilized to create our KG.

E.1 Support Statements

Here are a few examples of manually created support statements from which we reverse-engineered a list of relevant relations.

- Approximately 8 million metric tons of plastic waste enter the oceans every year, endangering marine life and ecosystems.
- Residential energy use accounts for about 20% of global CO2 emissions.
- Deforestation accounts for approximately 11% of global greenhouse gas emissions.
- The transportation sector is responsible for around 14% of global greenhouse gas emissions.
- The average food item in the United States travels about 1,500 miles to reach consumers.
- Approximately one-third of all food produced globally goes to waste.
- Energy-efficient appliances can reduce energy consumption by up to 50% compared to standard models.
- In 2020, renewable energy sources accounted for 29% of global electricity generation.
- Freshwater scarcity affects over 2 billion people worldwide.
- Approximately 1 million animal and plant species are currently threatened with extinction.

E.2 Selected Relations

Here is a list of the 46 relations we have chosen, along with sample triples that utilize them:

Relation	Example
HasStatistic	(Carbon Emissions, HasStatistic, 4.3 metric tons per capita)
HasNumericValue	(Global electricity generation, HasNumericValue, 90%)
HasUnitOfMeasurement	(Water Conservation, HasUnitOfMeasurement, gallons per household per month)
HasContext	(Climate Change, HasContext, global temperatures rising)
HasSource	(Environmental Report, HasSource, IPCC)
HasSubject	(Sustainable Agriculture, HasSubject, soil health)
HasAction	(Climate Change Mitigation Strategies, HasAction, Implementing Carbon Pricing)
HasAverageValue	(Vehicle Emissions Regulations, HasAverageValue, 100 grams per kilometer)
HasMinValue	(Waste Reduction Campaign, HasMinValue, 30% reduction in waste production)
HasMaxValue	(Green Building Standards, HasMaxValue, LEED Platinum certification)
HasImpact	(Renewable Energy Project, HasImpact, reducing carbon footprint)
HasPercentileValue	(Renewable Energy, HasPercentileValue, 28% of Electricity Generation)
HasTrend	(Climate Action Plan, HasTrend, increasing adoption rates)
HasComparison	(Renewable Energy Adoption, HasComparison, surpassing fossil fuel usage)
HasImpact	(Electric Vehicle Adoption, HasImpact, reducing air pollution)
HasCorrelation	(Wildlife Corridors, HasCorrelation, Biodiversity conservation)
Reduces	(Sustainable Transportation, Reduces, carbon emissions)
Saves	(Water Conservation Measures, Saves, freshwater resources)
Decreases	(Plastic Ban Policy, Decreases, plastic pollution)
Increases	(Renewable Energy Incentives, Increases, adoption rates)
EfficiencyOf	(Energy Efficiency of Appliances, EfficiencyOf, Energy Star certified products)
PercentageOf	(Water Management Facilities, PercentageOf, Over 15% Total Electricity Consumption)
RatioOfFrequencyOf	(Renewable Energy, RatioOfFrequencyOf, Renewable Energy Adoption)
RateOf	(Fashion industry, RateOf, 10% of Carbon Emmisions)
VolumeOf	(Volume of Water Used in Agriculture, VolumeOf, 70%)
EmissionOf	(CO2 Emissions, EmissionOf, transportation sector)
ConsumptionOf	(Energy Consumption, ConsumptionOf, residential buildings)
ImpactOf	(Biodiversity Conservation Efforts, ImpactOf, preserving habitats)
BenefitOf	(Renewable Energy Benefits, BenefitOf, reducing dependency on fossil fuels)
AdvantageOf	(Sustainable Agriculture, AdvantageOf, improving soil health)
DisadvantageOf	(Renewable Energy Adoption, DisadvantageOf, initial high installation costs)
RiskOf	(Climate Change, RiskOf, extreme weather events)
PreventionOf	(Carbon Capture Technology, PreventionOf, CO2 emissions)
ProtectionOf	(Protected Areas, ProtectionOf, endangered species)
PreservationOf	(Ecosystem Restoration Projects, PreservationOf, natural habitats)
ConservationOf	(Waste Management Programs, ConservationOf, landfill space)
RecoveryOf	(Ocean Cleanup Initiatives, RecoveryOf, marine ecosystems)
ManagementOf	(Forest Management Practices, ManagementOf, timber resources)
RegulationOf	(Environmental Regulations, RegulationOf, industrial emissions)
PolicyOf	(Climate Change Policy, PolicyOf, reducing carbon emissions)
InitiativeOf	(Sustainability Initiative, InitiativeOf, local government)
StrategyOf	(Environmental Strategy, StrategyOf, corporate sustainability)
AdaptationOf	(Climate Adaptation Plan, AdaptationOf, changing climate conditions)
MitigationOf	(Climate Mitigation Measures, MitigationOf, greenhouse gas emissions)
HasCapacity	("Wind Energy", "HasCapacity", "63 GW")
HasPolicyTarget	("European Union", "HasPolicyTarget", "40% renewable electricity by 2030")

Table 5: Examples of relations and their corresponding triples in curated knowledge graphs.